ETHAN CHANG

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RESEARCH INTERESTS

Computational neuroscience, NeuroAI, biologically inspired neural networks, astrocyte-informed machine learning, reinforcement learning, brain dynamics

EDUCATION

Washington University in St. Louis - *St. Louis, MO* *2025 - Present*
Ph.D. in Neuroscience

Johns Hopkins University - *Baltimore, MD* *2025 - Present (Concurrent)*
M.S. in Computer Science

University of Rochester - *Rochester, NY* *2021 - 2025*
B.A. in Mathematics and B.S. in Neuroscience, High Distinction. Minor in Psychology

RESEARCH EXPERIENCE

Washington University in St. Louis - *St. Louis, MO* *2025 - Present*
Doctoral Researcher, Brain Dynamics and Control Research Group and Papouin Lab

- Co-advised by ShiNung Ching, Ph.D. (Department of Electrical and Systems Engineering) and Thomas Papouin, Ph.D. (Department of Neuroscience)
- Investigating computational models that incorporate astrocyte-inspired reinforcement learning architectures for understanding neural computation and adaptive behavior.

University of Rochester - *Rochester, NY* *2021 - 2025*
Undergraduate Researcher, Nedergaard Lab (Maiken Nedergaard, M.D., D.M.Sc.)

- Conducted research on the role of aquaporin-4 localized on astrocytic endfeet in glymphatic transport dynamics.

PUBLICATIONS

- Gahn-Martinez D, Giannetto M, Chang E, Beam N, Pla V, Nedergaard M. Chronic intraventricular cannulation for the study of glymphatic transport. eNeuro. 2025. doi: 10.1523/ENEURO.0537-24.2025
- Giannetto M, Gomolka R, Gahn-Martinez D, Newbold E, Bork P, Chang E, Gresser M, Thompson T, Mori Y, Nedergaard M. Glymphatic fluid transport is suppressed by the AQP4 inhibitor AER-271. Glia. 2024. doi: 10.1002/glia.24515

PRESENTATIONS

- Chang E (presenter), Giannetto M, Agarwald I, Vento N, Gahn-Martinez D, Nedergaard M. Aquaporin-4 expression on glymphatic clearance routes and function. Schwartz Discover Scholar Showcase. 2024.
- Chang E (presenter), Giannetto M, Gahn-Martinez D, Nedergaard M. Aquaporin-4 expression and size-dependent solute movement in the brain. University of Rochester Undergraduate Research Exposition. 2024.
- Chang E (presenter), Barth RK. Isolation of hydrogen sulfide-producing bacteria from the environment. Department of Microbiology and Immunology Poster Session. 2023.
- Giannetto M, Gomolka R, Gahn-Martinez D, Newbold E, Bork P, Chang E (presenter), Gresser M, Thompson T, Mori Y, Nedergaard M. Glymphatic fluid transport is suppressed by the AQP4 inhibitor AER-271. University of Rochester Undergraduate Program in Biology and Medicine Poster Symposium. 2023.

AWARDS AND FUNDING

Washington University in St. Louis *2025 - Present*
NIH T32 Predoctoral Trainee, Neuroscience Training Program (T32NS121881)

University of Rochester *2024*
Schwartz Discover Grant

TECHNICAL SKILLS

Selected Programming/ML: Python, Java, PyTorch, NumPy, pandas, scikit-learn, matplotlib

TEACHING AND MENTORING

University of Rochester, Department of Brain and Cognitive Sciences *2023*
Teaching Assistant, NSCI 201P: Basic Neurobiology Lab
- Instructor: Renee Miller, Ph.D

Mentoring
- Audrey Jung, University of Rochester, Class of 2027
- Isha Agarwald, University of Rochester, Class of 2026
- Nick Vento, University of Rochester, Class of 2026

SERVICE AND LEADERSHIP

WashU Center for Career Engagement *2025 - Present*
Graduate Student Advisory Board Representative
- Provided graduate student input on career resources, especially for students interested in industry pathways.

WashU St. Louis Neuroscience Outreach *2025 - Present*
Student Volunteer
- Participated in public neuroscience outreach at the Amazing Brain Carnival, St. Louis Science Center SciFest.

University of Rochester Student Association *2021 - 2025*
Deputy Chair for Academic Affairs Committee
- Helped lead a team working on initiatives to improve the student academic experience and expand undergraduate research opportunities.

University of Rochester Office of Undergraduate Research *2023 - 2025*
Student Research Ambassador
- Served as a student contact for research involvement and participated in admissions panels.

Clinical and Emergency Service *2021 - 2024*
Strong Memorial Hospital; RC-MERT
- Emergency Department Clinical Support and Emergency Medical Technician roles in high-acuity care settings, including during the COVID-19 Omicron surge.

SELECTED COURSEWORK

AI/ML: Machine Learning; Advanced Machine Learning; Deep Learning; Natural Language Processing; Reasoning Under Uncertainty (Bayesian Networks, Probabilistic Inference, Decision Theory, MDPs)
Math/Statistics: Probability; Mathematical Statistics; Stochastic Processes; Linear Algebra; Numerical Analysis; Multidimensional Calculus; Point-Set Topology
CS/Systems: Algorithms; Data Structures; Software Engineering; Cloud Computing; Computer Architecture; Computer Organization

LANGUAGES

English (native), Mandarin Chinese (conversational)